Total number of pages: 1
Degree: B.Tech

Roll no.:........................
Semester: 3

End Semester Examination, Nov.-Dec. 2025

Course Code: ITMTC301/INMTC301/ITMTC04/EIMTC301

Course title: Optimization Principles and Techniques

Time: 3 hours

Maximum Marks. 50

Note: Attempt all the questions in the given order only. Missing data/information(if any), may be suitably assumed and mentioned in the answer.

Q. No.	Questions	Marks	CO
Q 1	Attempt any two parts of the following:		
1a	A farm is engaged in breeding pigs. The pigs are fed on various products grown on the farm. In view of the need to ensure certain nutrient constituents (X, Y and Z), it is necessary to buy two additional products say, A and B. One unit of product A contains 36 units of X, 3 units of Y and 20 units of Z. One unit of product B contains 6 units of X, 12 units of Y and 10 units of Z. The minimum requirement of X, Y, and Z is 108 units, 36 units, and 100 units, respectively. Product A costs Rs. 20 per unit and product B Rs. 40 per unit. Formulate the problem as LPP to minimize the total cost and solve the problem using graphical method.	5	CO1
1b	Use two-phase simplex method to solve the follwoing LPP. Max Z = $10x_1 + 20x_2$ subject to $2x_1 + x_2 = 1$ $x_1 + 2x_2 = 5$ $x_1, x_2 \geq 0.$	5	CO1
1c	Find all basic feasible solutions of the given LPP without using simplex method and choose that one which maximizes Z. Max Z = $2x_1 + 3x_2 + 4x_3 + 7x_4$ subject to $2x_1 + 3x_2 + x_3 + 4x_4 = 8$ $x_1 - 2x_2 + 6x_3 - 7x_4 = -3$ $x_1, x_2, x_3, x_4 \geq 0.$	5	CO1
Q 2	Attempt any two parts of the following:		
2a	The following table shows the information on the availability of supply to each warehouse, the requirement of each market and unit transportation cost (in Rs.) from each warehouse to each market. $\begin{array}{cccccc} & M1 & M2 & M3 & M4 & Supply \\ W1 & 6 & 3 & 5 & 4 & 22 \\ W2 & 5 & 9 & 2 & 7 & 15 \\ W3 & 5 & 7 & 8 & 6 & 8 \\ Demand & 7 & 12 & 17 & 9 & \end{array}$ The present transportation schedule is W_1 to M_2 : 12 units, W_1 to M_3 : 1 unit, W_1 to M_4 : 9 units, W_2 to M_3 : 15 units, W_3 to M_1 : 7 units, and W_3 to M_3 : 1 unit. Find the optimum transportation cost.	5	CO2
2b	We have to assign five jobs, I, II, III, IV, V to five workers A, B, C, D, and E. The time taken by different workers (in hours) in completing different jobs is given below: $\begin{array}{cccccc} & I & II & III & IV & V \\ A & 85 & 75 & 65 & 125 & 75 \\ B & 90 & 78 & 66 & 132 & 78 \\ C & 75 & 66 & 57 & 114 & 69 \\ D & 80 & 72 & 60 & 120 & 72 \\ E & 76 & 64 & 56 & 112 & 68 \end{array}$ Find the mimimum time (in hours) taken by the workers to complete all jobs.	5	CO2
2c	Consider the following Primal Programming Problem. Max $4x_1 + 3x_2$ subject to $x_1 + x_2 \leq 8$ $2x_1 + x_2 \leq 10$ $x_1, x_2 \geq 0$ a Convert this primal (P) into associated dual. Then solve the dual problem using graphical method. b Write down solution of dual LPP from final simplex table of primal problem.	5	CO2
Q 3	Attempt any two parts of the following:		
3a	Solve the following two-person zero-sum game using the dominance property: $\begin{array}{ccccc} & B_1 & B_2 & B_3 & B_4 \\ A_1 & 6 & 8 & 3 & 13 \\ A_2 & 4 & 1 & 5 & 3 \\ A_3 & 8 & 10 & 4 & 12 \\ A_4 & 3 & 6 & 7 & 12 \end{array}$	5	CO3
3b	Determine the optimal strategies for both the players and the value of the game whose pay-off matrix is given as follows: $\begin{pmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 6 \\ 4 & 1 \\ 2 & 2 \\ -5 & 0 \end{pmatrix}$	5	CO3
3c	Find the range of values of λ and μ that will generate the entry (2, 2) a saddle point for the following game: $\begin{array}{c\|c\|c\|c} & \text{Player B} & & \\ \text{Player A} & B_1 & B_2 & B_3 \\ A_1 & 3 & 5 & 8 \\ A_2 & 9 & 6 & \lambda \\ A_3 & 5 & \mu & 7 \end{array}$	5	CO3
Q 4	Attempt any two parts of the following:		
4a	Solve the following integer LPP by using the suitable method: Min Z = $-2x_1 - 3x_2$ subject to $2x_1 + 2x_2 \leq 7$ $0 \leq x_1 \leq 2$ $0 \leq x_2 \leq 2$ x_1, x_2 are integers.	5	CO4
4b	Solve the following mixed-integer programming problem: Max Z = $x_1 + x_2$ subject to $3x_1 + 2x_2 \leq 5$ $x_2 \leq 2$ $x_1, x_2 \geq 0$ x_1 is integer.	5	CO4
4c	Use Dual Simplex Method to solve the following LPP. Min Z = $x_1 + x_2$ subject to $2x_1 + x_2 \geq 2$ $-x_1 - x_2 \geq 1$ $x_1, x_2 \geq 0.$	5	CO4
Q 5	Attempt any two parts of the following:		
5a	Use the method of Lagrange multipliers and concept of Bordered Hessian Matrix to solve the following NLP: Optimize Z = $x_1^2 - 10x_1 + x_2^2 - 6x_2 + x_3^2 - 4x_3$ subject to $x_1 + x_2 + x_3 = 7,$ $x_1, x_2, x_3 \geq 0$	5	CO5
5b	Obtain the necessary and sufficient conditions for the optimum solution of the following NLP using the concept of Bordered Hessian Matrix: Optimize Z = $4x_1^2 + 2x_2^2 + x_3^2 - 4x_1x_2$ subject to $x_1 + x_2 + x_3 = 15$ $2x_1 - x_2 + 2x_3 = 20$ $x_1, x_2, x_3 \geq 0$	5	CO5
5c	Solve the following NLPP using KKT conditions: Optimize Z = $2x_1 + 3x_2 - (x_1^2 + x_2^2 + x_3^2)$ subject to $x_1 + x_2 \leq 1$ $2x_1 + 3x_2 \leq 6$ $x_1, x_2 \geq 0$	5	CO5